AI-powered Virtual Humans

LLM-Powered Cognitive Architectures

Module 1 · Chapter 02
LLMs, Working Memory and Prompting



Deep Learning and LLMs [9] [10]

The language engine inside our agent.

Deep learning

Neural networks learn patterns from data through multiple layers.

Large Language Models (LLMs)

Deep-learning models trained on large amounts of text to process and generate language.



Where did these models come from?

Attention Is All You Need [1]

The Transformer (2017)

Vaswani et al. introduced a model that connects words using attention.

Outcome

Better translations with less time spent training.

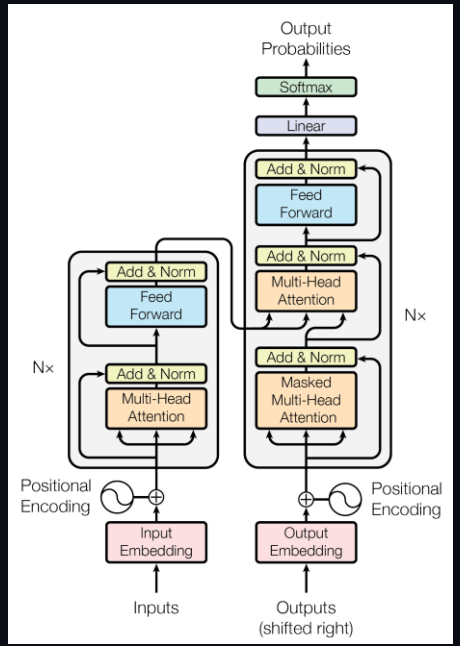


Figure 1 - The Transformer

GPT-2: Text Completion [2]

OpenAI (2019), 1.5B parameters

Predict the next token,
then repeat to continue the text.

Outcome

Coherent paragraphs, with errors and repetition still possible.

Prompt

The cat sat on the

Possible continuation

mat and fell asleep.

Illustrative example

Scaling Laws for Neural Language Models [3]

Kaplan et al., January 2020

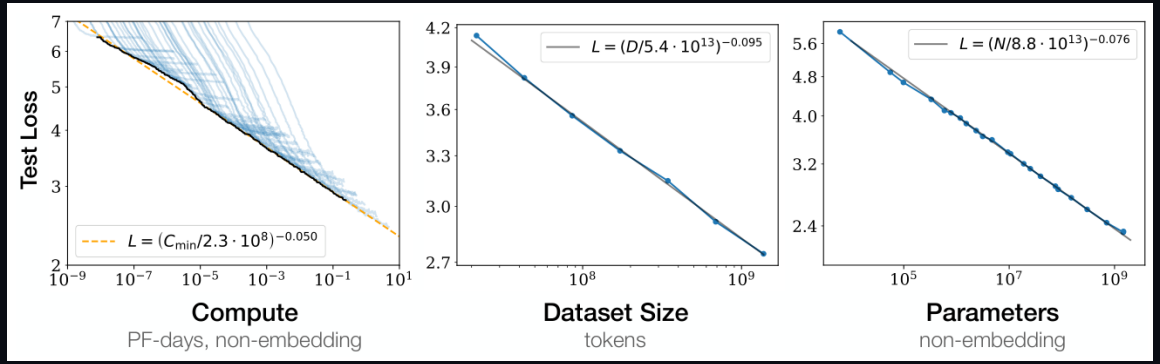
Size matters.

But what do we mean by bigger?

Scaling Laws for Neural Language Models [3]

Bigger wins…

More data, training computation and model parameters.



Outcome Lower prediction error as training resources increase.

GPT-3: Scaling Up [2] [4]

OpenAI, May 2020

GPT-3 applied the scaling strategy to a much larger language model.

Outcome

Stronger performance across many tasks using examples in the prompt.

GPT-2 (2019)	GPT-3 (2020)
1.5B	**175B**
parameters	parameters

≈117× as many parameters

Flipped Classroom

Read, check your understanding, discuss

Financial Times (2023) [5]

Generative AI exists because of the transformer

Open the interactive article

Individual reading

25 minutes
Note one question to discuss.

After reading

Kahoot, then questions and discussion.

Kahoot

14 questions to check your understanding

kahoot.it

Open the link or scan the QR.
Enter the game PIN shown on screen.



Language Models are Few-Shot Learners [4]

Brown et al. (2020) - GPT-3, 175B parameters

They evaluated language tasks using instructions and a few examples in the input.

Outcome

New cases handled without retraining the model for each task.

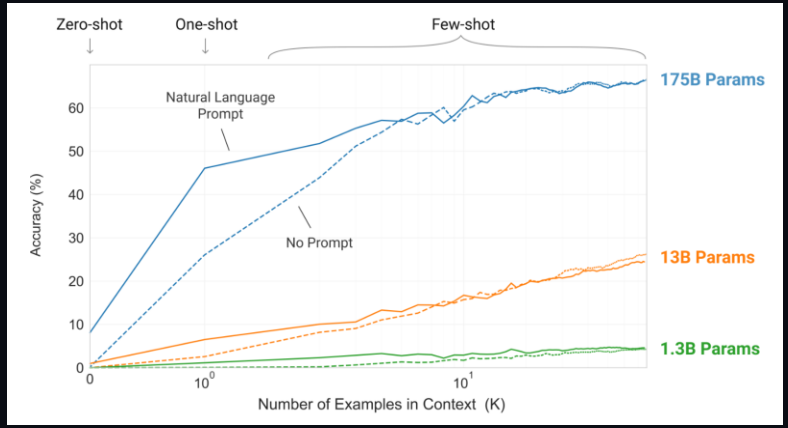


Figure 1.2 - examples supplied in context

In-Context Learning [4] [6]

The model can use examples in its prompt to infer a task pattern.



Context can also supply conversations, facts and memories.

No task-specific retraining is needed for this interaction.

Few-Shot Translation [4] [6]

Two examples, then a new sentence.

Prompt

English: Good morning.
Italian: Buongiorno.
English: Thank you.
Italian: Grazie.
English: Where is the entrance?
Italian:

Continuation ?

Few-Shot Translation [4] [6]

Two examples, then a new sentence.

Prompt

```
English: Good morning.
Italian: Buongiorno.
English: Thank you.
Italian: Grazie.
English: Where is the entrance?
Italian:
```

Continuation Dove si trova l'ingresso?

Completing a Question [4]

A question can also be the beginning of a text to continue.

Prompt

Question: What is the capital of France?
Answer:

Continuation Paris.

A Conversation Is a Prompt [4] [7]

Write a conversation and leave the next reply unfinished.

Prompt

```
Complete the conversation.
Visitor: Hello!
Guide: Welcome to the museum.
Visitor: Can you help me find the entrance?
Guide:
```

Continuation

Of course! Let me show you the way.

Adding Facts to the Prompt [8]

The application can supply information for the current reply.

Prompt

```
Complete the conversation.
Context: The guide is called PippoBot.
The current local time is 14:30.
Visitor: What is your name, and what time is it?
Guide:
```

Continuation I'm PippoBot. It is 14:30.

Bringing a Memory into the Prompt [8]

The program can retrieve a stored fact and include it in the input.

Long-term memory

Visitor #42
Prefers Italian



Prompt

Context: Visitor prefers Italian.
Visitor: Can we continue?
Guide:

Continuation Certo, continuiamo la visita.

The Context Window Is Limited [4] [8]

Only a finite amount of text can be available in one call.

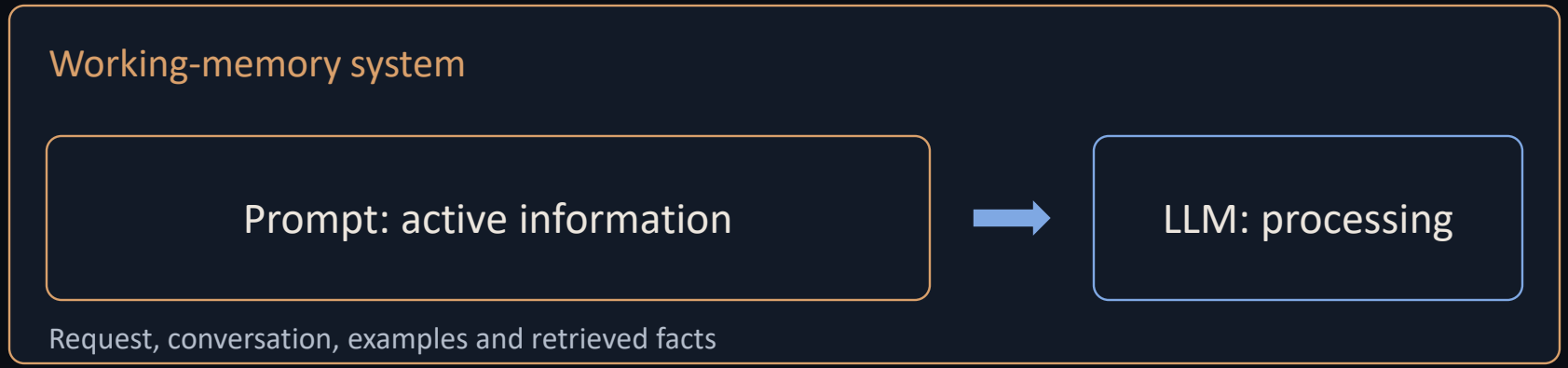


The application chooses what to keep, retrieve or summarise.

Like working memory, this is a limited active workspace.

Implementing Working Memory [8]

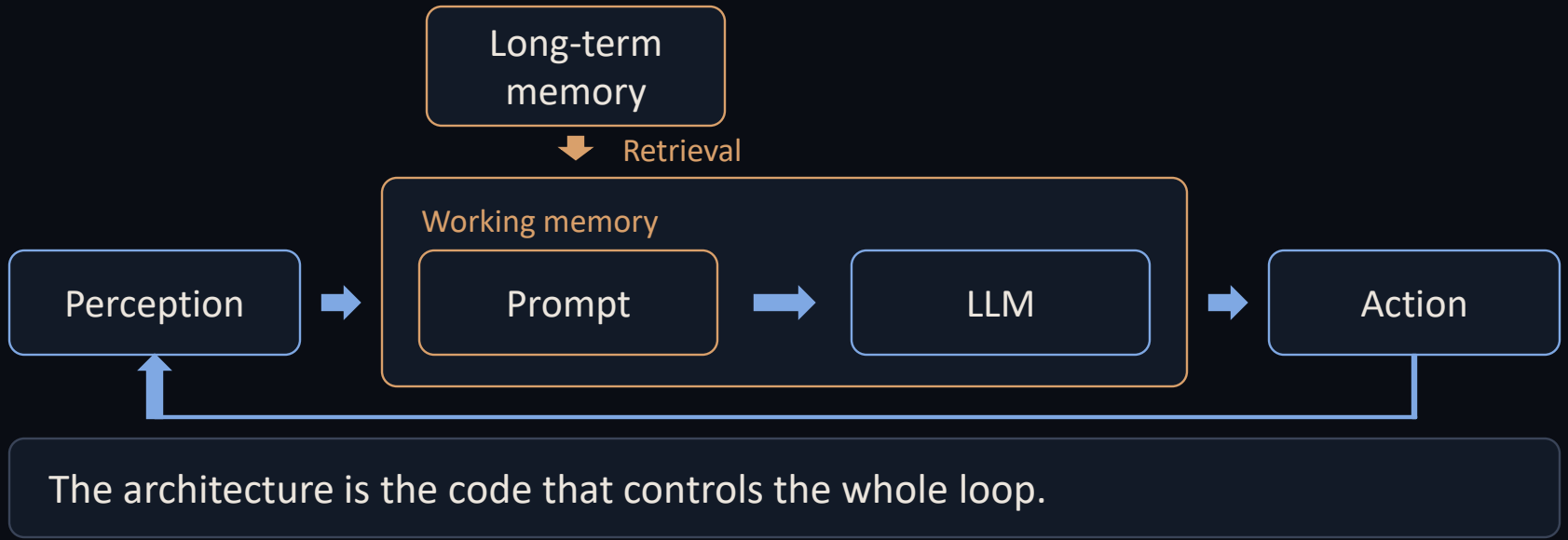
The code selects and updates the active context.



The code can be seen as a dynamic prompt manager.

A Computational Cognitive Architecture [8]

Code coordinates the LLM, working memory and other subsystems.



References

[1] Vaswani et al. (2017)

Attention Is All You Need

[2] Radford et al. (2019)

Language Models are Unsupervised Multitask Learners

[3] Kaplan et al. (2020)

Scaling Laws for Neural Language Models

[4] Brown et al. (2020)

Language Models are Few-Shot Learners

[5] Financial Times (2023)

Generative AI exists because of the transformer

[6] Dong et al. (2024 revision)

A Survey on In-context Learning

[7] Ouyang et al. (2022)

Training language models to follow instructions with human feedback

[8] Sumers et al. (2024)

Cognitive Architectures for Language Agents

[9] Goodfellow et al. (2016)

Deep Learning: Introduction

[10] Zhao et al.

A Survey of Large Language Models